

Lecture 28 — Cardinality

CS 173: Discrete Structures · Comparing and Counting Sizes of Sets

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1 Review and the Empty-Set Addendum

Cardinality comparisons must work for infinite sets too, so we phrase them in terms of functions, not counts.

Definition 1.1 (Cardinality comparison). For any sets A and B :

- (i) $|A| \leq |B| \iff \exists f (f : A \rightarrow B \wedge f \text{ injective})$.
- (ii) If $B \neq \emptyset$: $|A| \geq |B| \iff \exists f (f : A \rightarrow B \wedge f \text{ surjective})$.

Addendum (Empty target). We stipulate $|A| \geq |\emptyset|$ for all A .

Why the side condition? If $B = \emptyset$ and $A \neq \emptyset$, no function $A \rightarrow \emptyset$ exists, so the surjection definition would falsely give $|A| \not\geq |\emptyset|$. We patch this by fiat. The injection definition needs no such fix: the empty function $\emptyset \rightarrow B$ is vacuously injective, so $|\emptyset| \leq |B|$ comes for free.

2 Order-Theoretic Properties of $\leq$

Theorem 2.1 (Reflexivity). $\forall A (|A| = |A|)$.

Proof. $\text{id}_A : A \rightarrow A$ is a bijection. □

Theorem 2.2 (Duality). $\forall A \forall B (|A| \leq |B| \iff |B| \geq |A|)$.

Theorem 2.3 (Transitivity). $\forall A \forall B \forall C ((|A| \leq |B| \wedge |B| \leq |C|) \Rightarrow |A| \leq |C|)$.

Proof. If $f : A \rightarrow B$ and $g : B \rightarrow C$ are injective, so is $g \circ f$: from $g(f(a_1)) = g(f(a_2))$, injectivity of g then f yields $a_1 = a_2$. □

Theorem 2.4 (Antisymmetry — Cantor–Schröder–Bernstein). $\forall A \forall B ((|A| \leq |B| \wedge |A| \geq |B|) \Rightarrow |A| = |B|)$.

Remark. For finite sets, antisymmetry is immediate by counting; for infinite sets, a bijection has to be *built* from the two given functions. This is the content of Cantor–Schröder–Bernstein, proved separately.

Summary

Property	Statement
Reflexive	$ A = A $
Duality	$ A \leq B \iff B \geq A $
Transitive	$ A \leq B \wedge B \leq C \Rightarrow A \leq C $
Antisymmetric	$ A \leq B \wedge A \geq B \Rightarrow A = B $

Reflexivity, antisymmetry, and transitivity make $\leq$ a partial order on cardinalities. Duality just says $\geq$ is $\leq$ read backwards.

3 Inverses

To prove duality, we need: every injection has a surjective left inverse, and every surjection has an injective right inverse.

Theorem 3.1 (Left and right inverses). For any X, Y and $f : X \rightarrow Y$:

- (I) If f is injective, then $\exists g (g : Y \rightarrow X \wedge g \text{ surjective} \wedge g \circ f = \text{id}_X)$.
- (II) If f is surjective, then $\exists g (g : Y \rightarrow X \wedge g \text{ injective} \wedge f \circ g = \text{id}_Y)$.

The names “left” and “right” refer to where g sits in the composition. Applied to duality: an injection $A \rightarrow B$ produces a surjection $B \rightarrow A$ via (I), giving $|B| \geq |A|$; conversely (II) turns a surjection into an injection.

Caution (Empty-domain corner case). Part (I) fails as stated when $X = \emptyset$ and $Y \neq \emptyset$: the empty function $\emptyset \rightarrow Y$ is vacuously injective, but no $g : Y \rightarrow \emptyset$ exists. Either assume $X \neq \emptyset$ or handle the case separately, as with the empty-set addendum in Section 1.

The two-sided inverse

When f is bijective, both parts apply, and the resulting inverse is unique and itself a bijection.

Corollary 3.2 (Inverse of a bijection). If $f : X \rightarrow Y$ is a bijection, then

$$\exists! g (g : Y \rightarrow X \wedge g \text{ bijection} \wedge g \circ f = \text{id}_X \wedge f \circ g = \text{id}_Y).$$

Definition 3.1 (Inverse f^{-1}). The unique g above is called **the inverse** of f , written $f^{-1} : Y \rightarrow X$.

The jump from $\exists$ (one-sided) to $\exists!$ (two-sided) is real: a non-surjective injection has many left inverses, since values on $Y \setminus f(X)$ can be chosen freely. A bijection leaves no such freedom.

4 Categorical Vocabulary: Monic, Epic, Iso

The same three notions can be characterised purely by how f composes with other functions — no reference to elements.

Theorem 4.1 (Cancellation characterisations). For any X, Y and $f : X \rightarrow Y$:

- (1) f injective $\iff f$ is **monic**: $\forall W, \forall h_1, h_2 : W \rightarrow X, f \circ h_1 = f \circ h_2 \Rightarrow h_1 = h_2$.
- (2) f surjective $\iff f$ is **epic**: $\forall Z, \forall k_1, k_2 : Y \rightarrow Z, k_1 \circ f = k_2 \circ f \Rightarrow k_1 = k_2$.
- (3) f bijection $\iff f$ is an **isomorphism**: f has a two-sided inverse.

Remark (Why this matters). Mono- (one), epi- (onto), iso- (equal) match the elementary names. The point of the rephrasing: injective/surjective/bijective look *inside* X and Y ; monic/epic/iso only mention compositions. This generalises to settings where objects don't have “elements” — the entry point to category theory.

5 Finite Sets

So far “ $|X|$ ” has been a comparison device, not a number. For finite sets we can finally pin it down to a natural number.

Definition 5.1 (Finite set). X is **finite** iff $\exists n \in \mathbb{N} (|X| = |n|)$.

Here $n \in \mathbb{N}$ is identified with the set $\{0, 1, \dots, n-1\}$ (so $|n| = n$ in the everyday sense), and $|X| = |n|$ means there is a bijection $X \rightarrow n$.

Definition 5.2 (Cardinality of a finite set). If X is finite, define $|X| := n$, where $n \in \mathbb{N}$ is the unique natural number with $|X| = |n|$.

Remark (Why n is unique). Two distinct naturals $m \neq n$ satisfy $|m| \neq |n|$ (no bijection between sets of different sizes), so antisymmetry forces the n matching X to be unique. Without this fact, “ $|X| := n$ ” would be ill-defined.

Counting formulas

Theorem 5.1 (Subtraction for subsets). If X is finite and $Y \subseteq X$, then

$$|X \setminus Y| = |X| - |Y|.$$

Lemma 5.2 (Subtraction in general). If X is finite and Y is finite, then

$$|X \setminus Y| = |X| - |X \cap Y|.$$

Remark. The lemma reduces the general case to the previous theorem: $X \setminus Y = X \setminus (X \cap Y)$, and $X \cap Y \subseteq X$. The point is that only the part of Y *inside* X can be removed; elements of Y outside X were never there to begin with.

Closure properties

Theorem 5.3 (Intersections with finite sets). If X is finite and Y is a set, then $X \cap Y$ is finite.

Caution (Board check). The right-hand side of this theorem was partially obscured on the board. The natural reading — “ $X \cap Y$ is finite” — is the standard companion to the next theorem, but the conclusion could instead have been a counting formula such as $|X \cap Y| \leq |X|$. Worth confirming.

Theorem 5.4 (Subsets of finite sets are finite). If X is finite and $Y \subseteq X$, then Y is finite.
(★ exercise)

6 Counting Formulas

With finiteness defined, we can prove the elementary counting laws on products and unions.

Products

Theorem 6.1 (Cardinality of a product). If X and Y are finite, then

$$|X \times Y| = |X| \cdot |Y|.$$

Remark (Why the right-hand side makes sense). On the natural numbers, the product is the size of the standard set $n \cdot m = \{0, 1, 2, \dots, n \cdot m - 1\} = \{w \in \mathbb{N} \mid w < n \cdot m\}$, matching the von Neumann construction used to define $|n| = n$. So “ $|X| \cdot |Y|$ ” is genuinely a natural number, and the theorem says the product set has exactly that many elements.

Unions

Lemma 6.2 (Two-set inclusion–exclusion). If X and Y are finite, then

$$|X \cup Y| = |X| + |Y| - |X \cap Y|.$$

The correction $-|X \cap Y|$ accounts for elements counted twice when adding $|X|$ and $|Y|$ separately.

Theorem 6.3 (Inclusion/Exclusion — union bound). If $n \in \mathbb{N}$ and $A_1, A_2, \dots, A_n$ are finite sets, then

$$\left| \bigcup_{i=1}^n A_i \right| \leq \sum_{i=1}^n |A_i|.$$

Remark (Inequality, not equality). For n sets the exact formula has alternating-sign correction terms (the full inclusion–exclusion principle); the inequality stated here drops them and gives only the upper bound. Equality holds iff the A_i are pairwise disjoint, in which case all the correction terms vanish. The two-set lemma above is the smallest case where the correction is actually visible: $|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2| \leq |A_1| + |A_2|$.